

Jefferson David Kingston

Boston, MA | kingston.je@northeastern.edu | +1 857 284 3564 | linkedin.com/Jefferson-Kingston

EDUCATION

Northeastern University

Master of Science, Computer Science, GPA: 4.00/4.00

Relevant Coursework: Program Design Paradigm, Database Management Systems

Boston, MA

Sep 2025 – May 2027 (expected)

Vellore Institute of Technology

Bachelor of Technology in Computer Science and Engineering, CGPA: 8.77/10.00

Chennai, India

Jun 2021 – Jun 2025

EXPERIENCE

National University of Singapore (NUS)

Academic Intern

Singapore

Dec 2023 – Jan 2024

- Developed a sign-language interpretation system using MediaPipe Holistic for landmark extraction, OpenCV for frame preprocessing, and an LSTM model trained on gesture embeddings to achieve real-time classification.
- Constructed scalable TensorFlow training and validation workflows, optimized model latency using batching and quantization, and deployed the entire inference pipeline in a browser environment real-time user interaction.

AAM India Private Limited

Data Analyst Intern

Chennai

Oct 2023 – Nov 2023

- Assisted in data collection, preprocessing, and feature engineering for ML pipelines aimed at defect detection.
- Built baseline computer vision models for automated inspection and documented model performance metrics.

PROJECTS

Accurate Object Tracking for Aerial Marine Surveillance

Nov 2024

- Implemented YOLOv10 for detection and DeepSORT for tracking humans/ unusual activity via drone footage; focused on improving detection robustness across challenging marine conditions.

LSTM-Based Distributed DDoS Attack Detection

Apr 2024

- Built LSTM-based DDoS detection system processing multi-source network telemetry; achieved 94% detection accuracy with <50ms inference latency through online learning pipeline
- Designed an online learning with incremental streaming data ingestion and gradient updates, reducing model drift and improving model adaptability to evolving attack signatures without full retraining.

PATENTS & PUBLICATIONS

Intelligent Traffic Management System (Patent)

Oct 2024

- Published Indian patent (Application No: 202441075847) for an IoT-based traffic management system integrating YOLO, LiDAR speed estimation and acoustic sensors to prioritize emergency vehicles.

Multimodal Integration of Sensors with Computer Vision to Detect and Alert Wild Animal Intrusion

Apr 2026

- Co-developed a real-time wildlife intrusion detection system fusing YOLOv8n detection, DeepSORT multi-object tracking, and PIR/ultrasonic sensor validation, reducing false positives by 67% versus vision-only detection across a 14-class, 13,879-image dataset.
- Published in IEEE Xplore following oral presentation at RAEEUCCI-2026, SRM Institute of Science and Technology, Chennai (Apr 2026).

ACHIEVEMENTS

Winner Ideathon on “AI and Gen-AI” by SBB CFF FFS: Proposed a GenAI assistant to enhance SAP workflows

Finalist Dark Patterns Hackathon: used OCR and image recognition to flag misleading e-commerce content.

EXTRA-CURRICULAR

Dreadnaught Robotics- Programming team member; contributed to an ATV that placed 2nd at Hindustan ArcFest.

Church Media Team- Lead developer in building and maintaining the church website using React.js and MongoDB.

IEEE Photonics Club- Created visual content including posters and banners to support IEEE Photonics events.

TECHNICAL SKILLS

Technical: Python, C++, Java, C, JavaScript, SQL, OpenCV, TensorFlow, PyTorch, YOLOv10, DeepSORT, Code Llama, Streamlit, Docker, PostgreSQL, Git